High-Density Electromyography and Motor Skill Learning for Robust Long-Term Control of a 7-DoF Robot Arm

Mark Ison, Ivan Vujaklija, Bryan Whitsell, Dario Farina, *Senior Member, IEEE*, and Panagiotis Artemiadis

***Abstract*—Myoelectric control offers a direct interface between human intent and various robotic applications through recorded muscle activity. Traditional control schemes realize this interface through direct mapping or pattern recognition techniques. The former approach provides reliable control at the expense of functionality, while the latter increases functionality at the expense of long-term reliability. An alternative approach, using concepts of motor learning, provides session-independent simultaneous control, but previously relied on consistent electrode placement over biomechanically independent muscles. This paper extends the functionality and practicality of the motor learning-based approach, using high-density electrode grids and muscle synergy-inspired decomposition to generate control inputs with reduced constraints on electrode placement. The method is demonstrated via real-time simultaneous and proportional control of a 4-DoF myoelectric interface over multiple days. Subjects showed learning trends consistent with typical motor skill learning without requiring any retraining or recalibration between sessions. Moreover, they adjusted to physical constraints of a robot arm after learning the control in a constraint-free virtual interface, demonstrating robust control as they performed precision tasks. The results demonstrate the efficacy of the proposed man–machine interface as a viable alternative to conventional control schemes for myoelectric interfaces designed for long-term use.**

***Index Terms*—Electromyography (EMG), high-density electromyography (EMG), human–robot interaction, motor learning, myoelectric control, prosthetics, real-time systems, simultaneous control.**

I. Introduction

MYOELECTRIC control allows a convenient human–machine interface by transforming electromyography (EMG) signals into control outputs. Surface EMG (sEMG) provides non-invasive access to muscle activity. This type of control has been explored extensively due to its direct application to functional prostheses [1]–[3]. Recent extensions consider applications in other human–machine interfaces, such as robot teleoperation [4], [5], powered wheelchairs [6] and virtual joysticks [7]. However, transient changes in sEMG cause a trade-off between functionality (i.e., proportional and simultaneous control) [8], [9] and reliability (i.e. consistent long-term control without frequent retraining) [10], [11] in most control schemes.

Two control approaches are currently employed in conventional myoelectric control: direct control and pattern recognition (PR). Direct control links antagonistic muscles or muscle groups directly to a single degree-of-freedom (DoF) [12]. Various switching methods sequentially transition between different DoFs or functions in finite state machines [13]. These simplistic controllers provide reliable controls, but lack the functionality to smoothly operate multiple DoFs [14].

PR methods utilize classification [15] and regression [16] techniques to decode a mapping between myoelectric inputs and desired outputs [1]. This enhances functionality compared to direct controls by enabling multiple DoFs without explicit switching methods. However, increased functionality requires an updated training set which is highly dependent on the user's motion repeatability [15] and influenced by many external factors [17]–[20]. Thus, the decoding often overfits to a small set of the full input space, and performance tends to degrade over time [21]. Adaptive [21]–[23] and pre-trained models [24]–[26] attempt to avoid this effect with varied success.

An alternative method utilizes concepts of motor skill learning and brain plasticity to extend direct control principles to multiple DoFs [27]. Linear transformations between EMG inputs and control outputs naturally provide both simultaneous and proportional control [28]. The surjective mapping creates a redundancy in the control scheme which reduces the precision needed in muscle activations to control the entire task-space [29]. These methods have shown a user's ability to learn a given mapping, regardless of its intuitiveness [30], and to develop muscle synergies associated with enhanced control of the myoelectric interface [31], [32].

While these motor learning implementations consistently demonstrate session-independence, they rely on targeted electrode placement over biomechanically independent muscles to avoid biomechanical constraints [33] and to enhance learning potential [30], [32]. Robust extensions to three or more DoFs originally required a large number of input muscles [34] to maintain redundancy in the control scheme and to avoid unintended outputs [29]. However, Ison and Artemiadis [35]

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introduced a surjective mapping function enabling simultaneous control of 3-DoFs using four independent muscle inputs.

This work presents a novel myoelectric control scheme capable of real-time simultaneous and proportional control of a large number of DoFs, without requiring retraining between sessions, targeted electrode placement, nor biomechanically independent muscle inputs. A novel muscle synergy-inspired decomposition transforms high-density (HD) sEMG into independent activation signals, which linearly map to control outputs. The algorithm (detailed in Section III-A-2) extends the DoF-wise non-negative matrix factorization (NMF) algorithm proposed by Jiang *et al.* [36]. DoF-wise NMF extracts muscle synergies related to hand kinematics, and the extension transforms these synergies into independent activation signals to ensure the outputs can be operated autonomously for complete simultaneous and proportional control of the interface. The method is demonstrated using concurrent control of 4-DoFs in a finite state machine (FSM) to control a 7-DoF robot arm, and validated via learning trends and session-independent characteristics identified in other motor learning-based schemes [27]–[32]. To the best of the authors' knowledge, no other myoelectric control scheme has demonstrated real-time simultaneous and proportional control of 4-DoFs from untargeted sEMG of a single arm. Further, previous myoelectric control of a 7-DoF robot arm required retraining each session [37], [38].

II. Related Work

A. Motor Learning

Learning a new motor task typically involves a three stage learning process [39]. The initial cognitive stage carries a significant mental load as users explore and gather information relevant to the task. With repetition, users enter an associative stage with reduced conscious effort and increased performance. Finally, an autonomous stage is reached, where tasks seem unconscious and intuitive to perform [39].

These principles are associated with brain plasticity and commonly utilized in brain-machine interfaces [40]. Similar learning trends are seen with myoelectric interfaces. Users adapt to controls within a single session regardless of their initial intuitiveness or relationship with kinematics [29], [41]. Radhakrishnan *et al.*[27] found that this adaptation results in an exponential performance enhancement. Antuvan *et al.* [30] observed that less intuitive controls have a higher learning rate such that performance quickly converges with more intuitive control schemes. These adaptations are associated with the dynamic formation of new muscle synergies, resulting in more efficient control of the interface as users presumably enter the later stages of motor learning [31], [32]. Ison and Artemiadis found a correlation between these developing muscle synergies and a long-term learning component, both of which contribute to performance retention among similar tasks, efficient generalization to new tasks, and effective transfer to new interfaces [28], [31].

Clingman *et al.* [42] propose that these characteristics can be applied to myoelectric training systems for more robust interaction with prosthetic devices. Ison and Artemiadis [35] demonstrated the potential benefits of such a training system as subjects implicitly learned to control a robot arm by interfacing with an analogous virtual object via a non-intuitive control scheme. The subjects naturally transferred this learning when interacting with the physical robot, feeling the controls were intuitive despite them having no relation to human kinematics.

All of the above control schemes target biomechanically independent muscles as inputs. De Rugy, *et al.* [33] noted that dependent muscles may hinder motor learning and reduce performance while interacting with abstract myoelectric interfaces. The present work incorporates high density surface electromyography to eliminate this constraint and expand the capabilities of motor learning in myoelectric interfaces.

B. High-Density sEMG

sEMG signals are influenced by a variety of factors and transient changes which can render individual signals unreliable [1], [43], [44]. While many complications can be eliminated with invasive measures [45], they can be significantly reduced by recording a richer information set from HD electrodes.

PR-based myoelectric controls commonly use untargeted electrodes placed uniformly around the forearm [14], [46]. HD extensions use electrode grids to extract 2D information from muscle activity [47]. Muceli *et al.* [48] used these grids in a simultaneous 4-DoF control scheme, but only evaluated results offline. Tkach *et al.* [49] used similar grids to train a linear discriminant analysis classifier on arm and hand motions for subjects with amputations, concluding that the richer processing led to more robust control. Unfortunately, the increased number of signals bring enhanced concerns for overfitting, limiting such PR controls during long-term use.

Alternative control techniques with HD sEMG propose semi-unsupervised methods. Jiang *et al.* [36] introduced a muscle synergy-inspired extraction method for motions along desired DoFs in the control scheme, providing simultaneous control for an intuitive motor learning-like 2-DoF control scheme [50]. Muceli *et al.* [51] demonstrated robustness to different electrode number, shifts, and configurations. Although promising, these controls have yet to demonstrate session-independence for operation of more than 2-DoFs. This work adapts these methods to generate a session-independent control scheme capable of controlling more DoFs.

III. Methods

This experiment evaluates performance characteristics and control capabilities of a novel control scheme operating a 7-DoF myoelectric interface with 4-DoF simultaneous and proportional control. Three sessions are conducted on distinct days. In the first two sessions, a user learns the control scheme by interacting with a helicopter in virtual reality (VR) to complete a set of tasks. The final session involves controlling a KUKA Light Weight Robot 4 (LWR 4) and an attached Touch Bionics iLIMB Ultra to complete three precision tasks using the same control scheme. Subjects are split into two groups. The experimental group interacts with the two interfaces via HD sEMG inputs, while the control group uses keyboard inputs. The control group acts as a reference for learning using definitive, noiseless inputs on the non-intuitive control scheme.

A. Muscle Synergy-Inspired Decomposition

1) Muscle Synergy-Based Model: Jiang *et al.* [36] modeled HD sEMG observations, $\mathbf{Y}(t)$, as instantaneous mixtures of muscle activation signals, $\mathbf{F}(t)$. Following the notation in Muceli *et al.* [51], the relationship is defined as:

$$\mathbf{Y}(t) = \mathbf{W} \cdot \mathbf{F}(t) \tag{1}$$

where $\mathbf{W}$ is a channel weighting matrix representing the contribution of the activation signals to the signal received at each electrode. Each column $\mathbf{W}_i$ approximates a muscle synergy provided as high-level input by a user when interacting with myoelctric interfaces [1]. This model is proposed as the basis for extracting robust control signals from HD sEMG [51], and is similarly used as the basis for this control paradigm.

$\mathbf{W}$ is generated using the DoF-wise NMF algorithm described in [36]. This algorithm assumes that two activation signals control a single DoF on a joint, one in the positive direction, $f_i^+(t)$, and one in the negative, $f_i^-(t)$. Thus, $\mathbf{W}$ becomes a $n \times m$ matrix, where $m = 2d$, with n sEMG channels and d DoFs of interest [36]:

$$\mathbf{Y}(t) = \left[\mathbf{W}_1^+ \mathbf{W}_1^- \cdots \mathbf{W}_d^+ \mathbf{W}_d^-\right] \cdot \begin{bmatrix} f_1^+(t) \\ f_1^-(t) \\ \vdots \\ f_d^+(t) \\ f_d^-(t) \end{bmatrix} \tag{2}$$

2) Novel Activation Signal Extraction: This paper proposes a novel method to extract $k < m$ quasi-independent activation signals by approximating a subset of k independent columns in $\mathbf{W}$. If all $\mathbf{W}_i$ are orthonormal, $\mathbf{W}$ is a semi-orthogonal matrix satisfying $\mathbf{W}^T\mathbf{W} = I$, and the left inverse of $\mathbf{W}$ exists: $\mathbf{W}_{left}^{-1} = \mathbf{W}^T$. $\mathbf{Y}(t)$ can then be decomposed into independent activation signals:

$$\mathbf{F}(t) = \mathbf{W}^T \cdot \mathbf{Y}(t), \tag{3}$$

providing independent control inputs to the interface.

The DoF-wise NMF algorithm does not guarantee a semi-orthogonal $\mathbf{W}$. Moreover, the variability of sEMG causes uncertainty in the exact values of $\mathbf{W}$. The proposed algorithm produces a $n \times k$ semi-orthogonal matrix, $\hat{\mathbf{W}}$, decomposing sEMG into quasi-independent control inputs, $\hat{\mathbf{F}}(t)$, approximating activation signals $\mathbf{F}(t)$. Given the 4×4 Gaussian kernel $\mathbf{G}$, function $\delta(\mathbf{V})$ thresholding $\mathbf{V}$ to 0 at one standard deviation below the maximum element in $\mathbf{V}$, and 2D convolution operator $*$, the algorithm generating $\hat{\mathbf{W}}$ is as follows:

1) Rearrange all $\mathbf{W}_i$ according to the 2D configuration of the HD electrode grid.
2) $\forall \mathbf{W}_i (i \in \{1..m\}) : W_i' = \delta(\mathbf{W}_i) * \mathbf{G}$.
3) $m - k$ times do: Merge $\mathbf{W}_a'$ and $\mathbf{W}_b'$, where $\mathbf{W}_a'$ and $\mathbf{W}_b'$ have the closest cosine similarity of all $\mathbf{W}_i'$ pairs.
4) $\forall \mathbf{W}_i' (i \in \{1..k\}) : \mathbf{W}_i' = \delta(\mathbf{W}_i') * \mathbf{G}$.
5) $\forall \mathbf{W}_i' : \hat{\mathbf{W}}_i = \frac{\mathbf{W}_i'}{|\mathbf{W}_i'|}$, reshaped to a row vector.

Fig. 1 demonstrates the algorithm visually. $\hat{\mathbf{W}}$, with Gaussian blurred, orthonormal columns, satisfies (3) and forms the basis of a session-independent control scheme. The mixture of Gaussians, represented by $\hat{\mathbf{W}}$, act as a spatial low-pass filter on the noisy EMG input signals. The sparser $\hat{\mathbf{W}}$ also reduces the influence of cross-talk for easier isolation of control outputs and enhanced simultaneous control. The consistency of the resulting decomposition is a function of overlap between the approximated activation signals and the user's true underlying activation points. By only considering independent activation points, the robustness is a function of electrode span within and electrode distance between each activation signal versus potential electrode displacement.

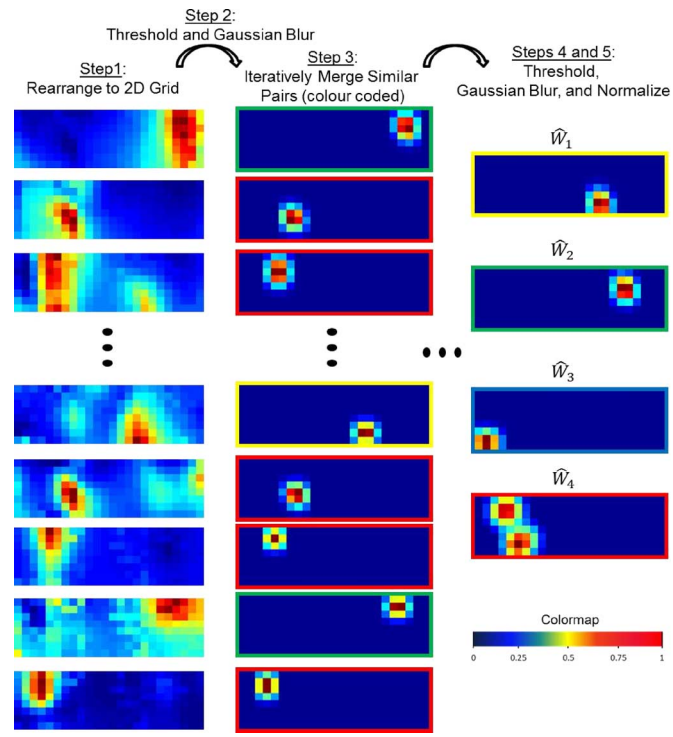
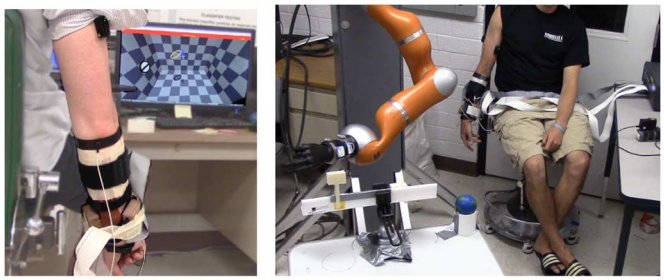


Fig. 1. Summary of the proposed 5-step algorithm to generate $\hat{\mathbf{W}}$ with four quasi-independent control inputs from a noisy $\mathbf{W}$. Each column of both $\mathbf{W}$ and $\hat{\mathbf{W}}$ is represented with elements rearranged according to their topographic position on high-density electrode grids, as described in Fig. 3 below.

B. Experimental Setup

The VR and robot control setups for the experimental group are shown in Fig. 2. HD sEMG signals are recorded from the right forearm muscles, approximately two inches below the elbow at the ulnar bone, using three equally spaced semi-disposable adhesive 8×8 electrode grids, with 10 mm inter-electrode distance. In contrast to previous studies involving multiple sessions with untargeted electrodes [21], exact electrode positioning was not marked in this study to simulate cross-session performance in a realistic use scenario. The skin is cleansed with water and a reference electrode is placed on the subject's elbow.

The 192 monopolar signals are acquired using EMG-USB2, OT Bioelettronica amplifier with gain set to 1000, bandpass filtered at 3–900 Hz, sampled at 2048 Hz with 12-bit A/D conversion, and broadcast via TCP for further processing, as in [51]. Two additional sEMG signals are recorded on the biceps brachii



(a) (b)

Fig. 2. Experimental setup with HD sEMG electrodes. (a) Session 1 and 2: VR interface controlling a helicopter's position, orientation, and color. (b) Session 3: LWR 4, iLIMB, and three target objects to grasp and move based on controls learned in the VR interface.

TABLE I
FINITE STATE MACHINE CONTROL AXES

Control Axis	Position State	Orientation State
1	X	Yaw (ϕ)
2	Y	Pitch (θ)
3	Z	Roll (ρ)
4	Color (VR) or Hand Open/Close (Robot)	

(BB) and triceps brachii (TB) using wireless electrodes (Delsys Trigno Wireless, Delsys Inc.). These bipolar electrodes are acquired with a gain of 500, digitized with 16-bit depth at a frequency of 1926 Hz and broadcast via TCP. Both signals are received by a custom program using C^{++} and OpenGL API [52] to control both interfaces. The sEMG signals are processed in real-time and converted to control variables of the virtual helicopter (session 1 and 2) or robotic hand (session 3). The helicopter and hand respond to the outputs at 200 Hz.

C. Control Paradigm

The 7-DoF control scheme is implemented as a two-state finite state machine, with each state offering simultaneous control of velocities over 4-DoFs (see Table I). Control Axes 1–3 switch between controlling the global position and local orientation of the object. Co-contracting both BB and TB above a preset threshold induces the switch. Control Axis 4 is constant among both states, controlling the color of the helicopter and hand opening/closing of the robot.

The input sEMG is pre-processed to provide linear envelopes as input, $\mathbf{Y}(t)$. The mapping function is adapted from [31], with a linear mapping between the 194×1 vector $\mathbf{Y}(t)$ $(3*64+2)$ and 4×1 vector $\mathbf{U}(t)$ of control outputs:

$$\mathbf{U}(t) = g\mathbf{M} \cdot \hat{\mathbf{W}}^{\mathbf{T}} \cdot [(\mathbf{Y}(t) - \sigma) \circ u(\mathbf{Y}(t) - \sigma)], \quad (4)$$

where $\circ$ is an element-wise matrix multiplication, $u(*)$ is the unit step function, σ is the muscle activation threshold, and g is the output gain. $\hat{\mathbf{W}}$ is the muscle synergy inspired 194×6 decomposition matrix reducing $\mathbf{Y}(t)$ to quasi-independent control inputs $\hat{\mathbf{F}}(t)$, as described previously. $\mathbf{M}$ is a semi-random mixing matrix converting $\hat{\mathbf{F}}(t)$ to the control outputs $\mathbf{U}(t)$. The resultant $\mathbf{U}(t)$ is averaged over the last five outputs to reduce effects of any motion artifacts, electrode disconnections, or unintended muscle twitches. Both σ and g can be tuned for each subject, but in general, $\sigma = 0.01$ mV prevents undesired outputs from resting muscles and $g = 50$ provides a conservative sensitivity trade-off between too much and too little muscle activation required for movement.

1) Pre-Processing: Both sets of sEMG inputs are pre-processed to provide linear envelopes to the control scheme. The HD sEMG signals are first subtracted from the mean of all signals to remove external common noise, and then rectified and low-pass filtered (fourth-order zero-lag Butterworth filter, cut-off frequency 3 Hz). Finally, the signals are filtered by a 3×3 median filter to minimize the effects of any local disturbances. The additional TB and BB signals are rectified, lowpass filtered (fourth-order zero-lag Butterworth filter, cut-off frequency 3 Hz), and normalized with respect to the subject's maximal voluntary contraction (MVC), as found during the initial calibration. Both sets are sub-sampled to 200 Hz and merged to $\mathbf{Y}(t)$, with TB and BB as the last elements.

2) Calibration: An initial calibration phase generates the session-independent $\hat{\mathbf{W}}$ unique for every subject, as described in Section III-A. With 192 HD electrodes spanning the circumference of the forearm, eight DoFs are considered here—four coarse wrist motions (wrist flexion/extension, wrist pronation/supination, ulnar/radial deviation, hand open/close) and four fine finger motions (flexion/extension of the index, middle, ring, and pinkie fingers). The calibration data was collected following the procedure described in [47]. Subjects were prompted by a monitor to move along each direction at a pace of roughly three seconds per motion. Each movement was repeated four times, summing to a total of 64 three-second recordings across the sixteen listed motions (half-DoFs) used to initialize a 192×16 $\mathbf{W}$.

Because the TB and BB are not part of the HD grid, their contribution is ignored until after the initial calculation of $\hat{\mathbf{W}}$ with $k = 4$. The contribution of both TB and BB are then appended to the fifth and sixth columns of $\hat{\mathbf{W}}$ at the 193rd and 194th row, respectively, with the remaining elements in those rows and columns set to zero. Thus, the 194×6 $\hat{\mathbf{W}}$ has the first four control inputs decomposed from forearm muscles (see Fig. 3), and the final two coming from TB and BB.

During this calibration, subjects also performed their MVC for TB and BB to set the switching threshold at 50% of it.

3) Mixing Matrix: To generate a control scheme capable of spanning the entire task space with minimal inputs, the mixing matrix, $\mathbf{M}$, is a random matrix optimized with a cost function maximizing the angles between row vectors and subject to the following constraints, where $\mathbf{M}_i$ is a column vector in $\mathbf{M}$:

1) $\mathbf{M}_5 = [0, 0, 0, 1]^T$ and $\mathbf{M}_6 = -\mathbf{M}_5$ to disconnect grasping from motion.
2) $\forall \mathbf{M}_i : |\mathbf{M}_i| = 1$ for equal contribution from all inputs.
3) All row vectors are zero mean to prevent motion at rest. For this experiment, $\mathbf{M}$ is as follows (see Fig. 4):

$$\mathbf{M} = \begin{bmatrix} 0.52 & -0.94 & 0.42 & 0.00 & 0.00 & 0.00 \\ 0.79 & 0.06 & -0.85 & 0.00 & 0.00 & 0.00 \\ -0.33 & -0.34 & -0.33 & 1.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.00 & 1.00 & -1.00 \end{bmatrix} \quad (5)$$

4) Robot Control: The robot control runs slightly different than the controls within VR due to joint limits, singularities, and torques. LWR 4 operates in Cartesian impedance control using inverse kinematics when the control state is in position, and joint impedance control using forward kinematics of the

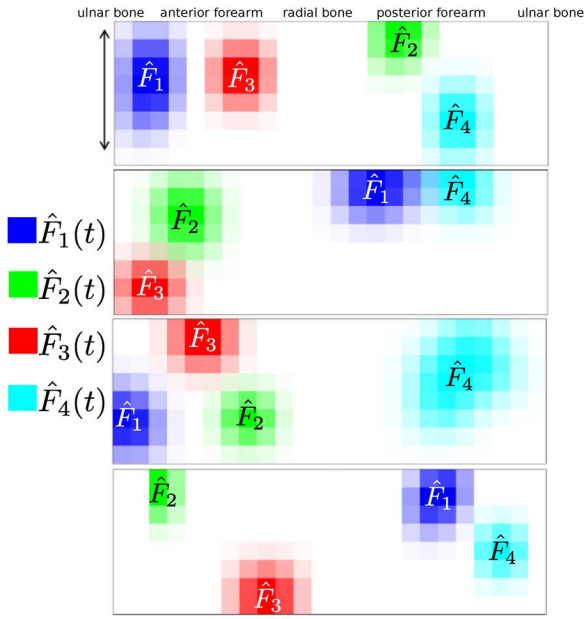


Fig. 3. Visualization of the first four columns of the muscle-synergy inspired decomposition matrix $\hat{\mathbf{W}}$, transforming high density sEMG $\mathbf{Y}(t)$ to four of the six quasi-independent control inputs $\hat{\mathbf{F}}(t)$. Each column $\hat{\mathbf{W}}_i$ is represented by a color, with elements rearranged according to their topographic position on the HD electrode grids. The intensity of the color indicates the weight of the elements in $\hat{\mathbf{W}}_i$. prox: proximal, dist: distal. This representation shows the three 8×8 grids as if they were contiguous around the circumference of the arm. In reality, each grid is separated by some distance depending on the subject's forearm circumference.

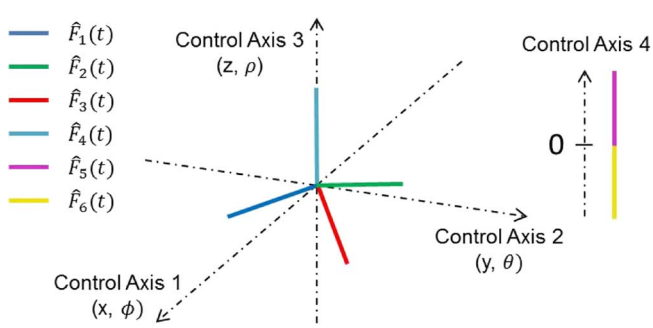


Fig. 4. Visualization of mapping $\mathbf{M}$, transforming control inputs $\hat{\mathbf{F}}(t)$ to four output control axes $\mathbf{U}(t)$, where each axis is as defined in Table I.

wrist joints when the control state is in orientation mode. The switch to joint impedance reduces the risk of exceeding joint velocity and position limits while rotating through singularities in the null space. Global ρ, ϕ, and θ are limited to $\pm(\pi/3)$ to avoid physical joint limitations while rotating. The iLIMB is controlled via velocity commands to all fingers over Bluetooth.

5) Noiseless Keyboard Interface: The control group uses keyboard inputs, with US layout, as $\hat{\mathbf{F}}(t)$, with simultaneous control offered by pressing multiple keys:

$$\mathbf{U}(t) = g\mathbf{M} \cdot \hat{\mathbf{F}}, \tag{6}$$

where $\mathbf{U}$, g, and $\mathbf{M}$ are defined as in (4), and visualized in Fig. 4. Subjects control color changes/hand grasping with their left hand ($\hat{F}_{5,6} = \{1, 2\}$) and position/orientation with their right hand ($\hat{F}_{1,2,3,4} = \{j, k, l, ;\}$) without needing to move

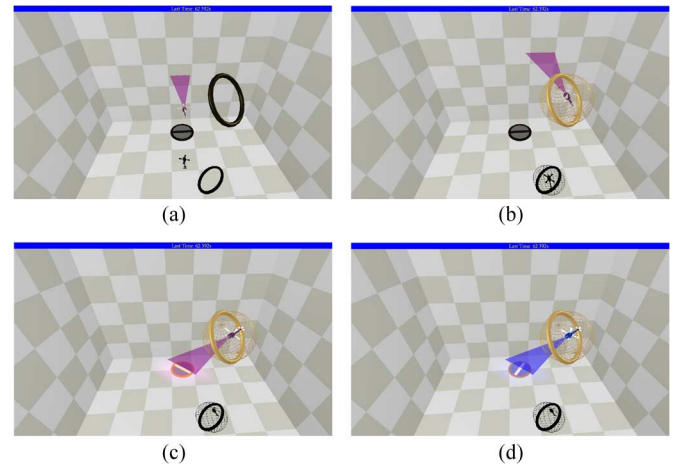


Fig. 5. Subtask sequence for VR. The helicopter (a) starts from an initial configuration, (b) moves to the center of the ring, (c) rotates to point toward the target on the wall, (d) changes to the color shown in the top bar cue. The color task can be completed simultaneously, but (b) and (c) must be completed in order. Once all three subtasks are complete, the helicopter resets, and new ring, target, and color are shown after a 10 second rest.

their fingers off the keys. The magnitude of $\hat{F}$ was given by 90% of the maximum synergy input used by all subjects in the experimental group, allowing them to simulate proportional control by tapping the keys. This provides an ideal scenario for learning as subjects interact with the non-intuitive controls.

D. Experimental Protocol

Subjects interact with the control scheme over three sessions on distinct days. None of the subjects were initially aware how to control the interface, although they were shown how to switch between control states. They were asked to learn the simultaneous controls simply by interacting with them. The experimental group used sEMG inputs as described above, with an initial calibration phase before the first session to generate $\hat{\mathbf{W}}$ associated with the subject's underlying muscle anatomy. The control group used keyboard inputs as a noiseless substitute for sEMG.

1) Session 1: This session introduced the VR and its controls to the subject. All subjects were initially given 20 minutes to become familiar with the controls, in which example tasks were presented without requiring completion. Subjects were encouraged to explore the interface and understand how to control the helicopter along all 7 available DoFs. This time presumably encompasses the initial exponential learning documented previously [27], [29], [30].

After 20 minutes, subjects were asked to complete 26 tasks. At this point, it is expected that they will show more linear learning trends [31]. Each task consisted of three subtasks exercising control across all DoFs (see Fig. 5):

1) Move the helicopter to a ring displayed on the screen.
2) Rotate the helicopter toward a target displayed on a wall.
3) Change the helicopter's color to match the top bar cue.

Once the subject completed the full task, the helicopter returned to an initial position, orientation and color, and a ten second break was given. There was no task time limit to encourage exploration and reduce any effect of einstellung (i.e. becoming stuck in a non-optimal solution) while learning [53].

(a) (b) (c)

Fig. 6. Subtask sequence for the robot interface. The robot hand is controlled to grasp two clothespins (a and b, subject perspective) and a ball (c, top view). The object is then placed into the bin below the table. The order in which these tasks are completed is determined by each subject. (a) Clothespin 1 (b) Clothespin 2 (c) Ball.

In total, subjects were presented 26 distinct rings uniformly placed on a sphere surrounding the starting point of the helicopter. 26 distinct orientation targets were also presented, uniformly spread on the walls along the front half of the VR space. Each trial presented a random combination of ring position, target position, and prompted color such that after 26 trials each ring and target position, respectively, were displayed exactly once. Thus, completing the full set on the first session ensures that the subject can move in the full task space. The random arrangement of ring, targets, and colors were constant for each subject to ensure a consistent learning environment.

2) Session 2: This session occured at least 24 hours after session 1. Subjects were asked to complete as many VR tasks as possible for one hour to evaluate learning retention. Subjects were not given any exploration time nor assistance during the interaction, and use the same control scheme and $\hat{\mathbf{W}}$ calculated during session 1. Each cycle of 26 tasks presented a random combination of the complete sets of rings, targets, and colors from Session 1, held constant across subjects.

3) Session 3: The final session occurred between one and eight days after session 2, and introduced the robot interface while still using the same control scheme and $\hat{\mathbf{W}}$ calculated during session 1. Subjects were asked to complete three precision tasks by sequentially grasping a tennis-sized ball and two customized clothespins to place in a bin. The clothespins were 3D printed extensions of conventional ones to provide a larger gripping area. The 2 × 1.25 inch grasping pad allowed two iLIMB digits to close on the area for a more stable grasp

Fig. 2(b) shows the setup of the objects, with ideal grasps shown in Fig. 6. Subjects were required to activate all 7 DoFs with centimeter precision to grasp each object. The experiment was complete after grasping each object in any order.

E. Data Analysis

The first two sessions are split into eight equal blocks containing data from 25% of a single session, and performance metrics are compared among the control and experimental groups to determine the presence of motor learning. The final session records total completion time to indicate precision performance capabilities and any influential factors.

1) Learning Trends: During the first two sessions, performance is evaluated according to three metrics: completion time, throughput and path efficiency [54], [55]. Completion time, CT, is the time taken to successfully complete the task. Throughput, TP, measures both speed and accuracy by considering CT with respect to task difficulty, measured in bits/second according to Fitts' law standards [56]. Path efficiency, PE, is the ratio between the optimal distance, D, to complete the task and the actual path taken to reach the target [54].. The index of difficulty, ID, of a task is given by the Shannon Formulation [56]:

$$ID = \log_2\left(\frac{D}{W_D} + 1\right) \tag{7}$$

where D is the optimal distance to complete the task and W_D is the weighted cumulative error tolerance of all targets, held constant throughout this experiment. D is formulated similar to [41]:

$$D = \frac{1}{g}\sqrt{(\lambda\gamma_1 + \gamma_2)^2 + \gamma_3^2} \tag{8}$$

where g is as defined in (4), γ_1 is the straight line distance from the starting position to the center of the ring, γ_2 is the angular distance between the starting orientation of the helicopter and the target orientation with respect to the center of the ring, γ_3 is the internal distance from starting color to desired color, and $\lambda = 0.471$ is the ratio between output angular velocity in orientation state and output velocity in position state for equal input $\hat{\mathbf{F}}(t)$. $gamma_1$ and $gamma_2$ are additive because the position and orientation subtasks are sequential, while color can be performed simultaneously. TP is then:

$$TP = \frac{ID}{CT} \tag{9}$$

The three metrics CT, TP and PE are evaluated with respect to block number b to form a mean model of the learning curve. Based on [31], the initial 20 minute exploration time presumably encapsulates the initial exponential learning component. Thus, the metrics are assumed to contain approximately linear trends, and are fit to first degree polynomials:

$$CT(b) = \kappa_{ct} - \beta_{ct} b \tag{10}$$

$$TP(b) = \kappa_{tp} + \beta_{tp} b \tag{11}$$

$$PE(b) = \kappa_{pe} + \beta_{pe} b \tag{12}$$

where b represents the overall block number in session 1 and 2, κ represents initial performance, and β represents the learning rate for each component. For all metrics, a positive β indicates better performance and a significant learning component.

2) Robot Control: In the third session, subjects are evaluated on how quickly they perform all three precision tasks with the robot. This may be influenced by a number of immeasurable factors such as strategy and adaptation to the physical constraints of the robot. It may also be influenced by measurable factors such as performance in the virtual tasks, time between session 2 and session 3, and, for the experimental group, the choice of $\hat{\mathbf{W}}$. With a small number of data points, it is not expected to find a valid model of these factors to predict the final completion time. Thus, correlation coefficients between these factors and the final completion time identify relationships and rank the importance of each factor.

IV. Results

Eight healthy subjects (all male, age 19–40, 1 left-, 7 right-handed, forearm circumference 10–12 in, forearm length

9.75–11.25 in) formed the experimental group. Five additional subjects (all male, age 20–25) served as the control group. All subjects gave informed consent to the procedures as approved by the ASU IRB (Protocol: #1201007252).

Of the eight subjects in the experimental group, two subjects displayed outlier tendencies. One subject experienced sudden confusion during the second session (block 6), which caused a loss of control. This subject struggled to regain control throughout the remaining experiment. In contrast, another participant learned the controls almost entirely in the 20 minute exploration stage and performed significantly better than any other subject, even those in the control group, throughout all sessions. The results include both subjects, most noticeably through the learning trend inconsistencies at block 6.

A. Learning Trends

The average time between session 1 and 2 was 30 hours for the experimental group, and 25 hours for the control group. All but one subject found the controls easier at the start of the session 2, consistent with motor learning characteristics [57].

Completion times for both groups follow a roughly linear trend throughout sessions 1 and 2 (see Fig. 7 top). The fit for the experimental group reveals a significant learning rate despite the non-intuitive control scheme resulting in initial poor performance (see Table II first row). The control group also shows a significant learning rate, though significantly less than the experimental group, as subjects found the noiseless keyboard inputs easier to explore and quicker to learn. Consistent with previous studies [27], [30], the two groups quickly trend toward similar performance despite the initial gap.

The two groups similarly show significant learning trends in throughput (see Fig. 7 middle). The learning rates for both the experimental group and control group are not significantly different (see Table II row 2), although they are separated by an initial performance gap relating to the ease of discovering the appropriate control inputs.

Both groups also show significant learning in path efficiency (see Fig. 7 bottom). The learning rates for both the experimental group and control group are not significantly different (see Table II row 3), although they are separated by the initial performance gap, similar to throughput. The poor fit metrics are expected due to the bias toward higher variance as the mean path efficiency increases [31].

Both groups maintained the learning trends despite the break between sessions, as shown by block 4 and 5 in Fig. 7.

B. Robot Control

The average time between session 2 and 3 was 97 hours (∼4 days) for the experimental group, and 119 hours (∼5 days) for the control group. All but one subject found the controls consistent during the start of the third session, although all subjects in both groups perceived occasional delays in the control outputs, caused by generating outputs exceeding physical joint and velocity limits. An example grasp sequence is shown in Fig. 8. A supplementary video demonstrating the various precision tasks is available at: http://youtu.be/Qrel34jA4TQ.

On average, the control group finished all three tasks in 14.4 minutes (95% CI [10.8, 18.0]), while the experimental group finished in 30.6 minutes (95% CI [18.0, 43.1]). Although significantly higher on a student paired t-test ($p = 0.04$), the relative completion times, with the experimental group finishing twice as long as the control group, are expected considering the trends in Fig. 7. At the end of session 2, the experimental group finished tasks three times slower than the control group, but was trending toward equal performance.

The relationship between the final completion time and identified sources of influence are considered by correlation coefficients between the metrics for each subject, displayed

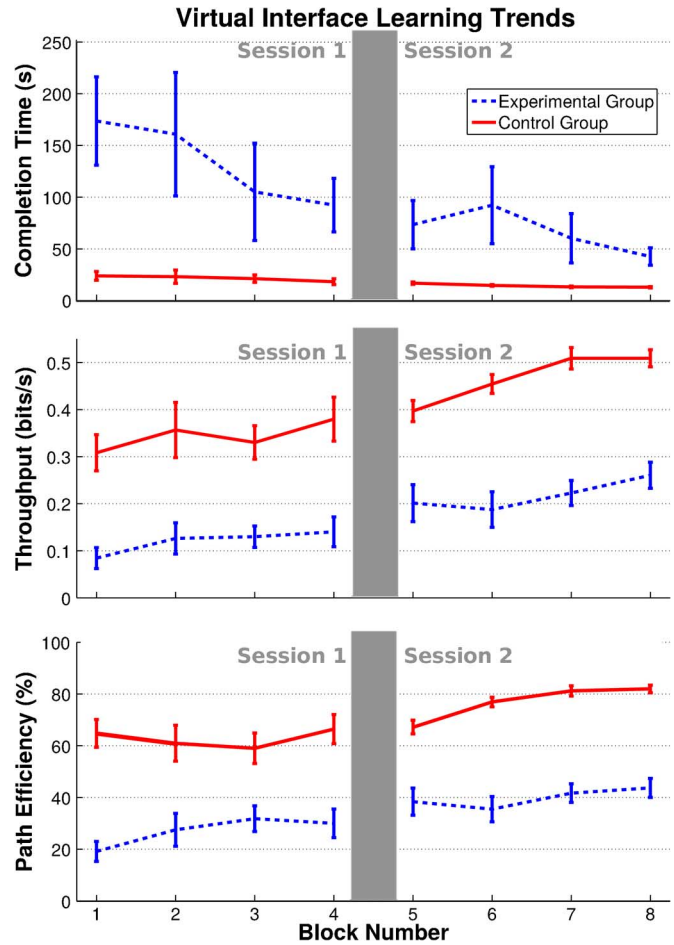
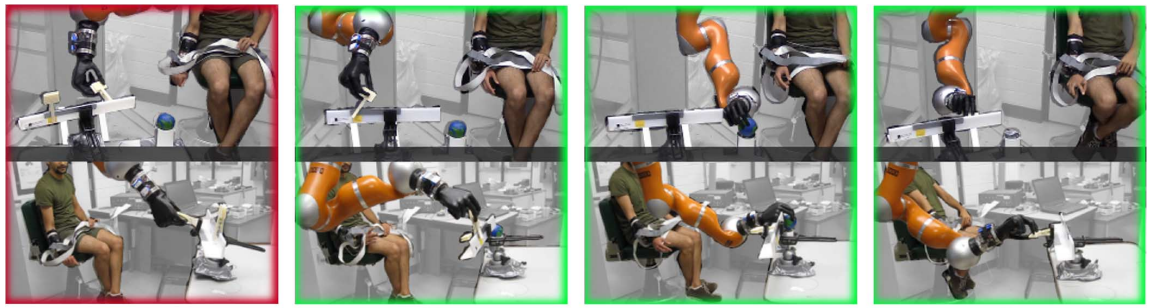


Fig. 7. Performance metrics as a function of block number for all subjects. The error-bars represent a 95% confidence interval within each block. Both groups display consistent improvement despite at least 24 hours between each session. With respect to completion times, the experimental group shows worse initial performance but a faster learning rate such that completion times are quickly converging. Both groups have similar learning rates in throughput and path efficiency, with an offset indicative of the additional exploration time needed for sEMG controls.

TABLE II
LEARNING TRENDS FITTING PARAMETERS

Metric	Group	β	β [95% CI]	κ	R^2
$CT(b)$	exp	17.10	[12.40,21.70]	177.0	0.94
	cont	1.66	[1.39, 1.92]	25.3	0.98
$TP(b)$	exp	0.024	[0.019, 0.029]	0.06	0.98
	cont	0.033	[0.028, 0.038]	0.26	0.97
$PE(b)$	exp	0.031	[0.024, 0.038]	0.20	0.10
	cont	0.037	[0.032, 0.042]	0.53	0.23

*exp: experimental group; cont: control group



(a) (b) (c) (d)

Fig. 8. Example chronological task sequence completed by a subject, with an example of an unsuccessful grasp (a) in red, followed by three successful grasps (b, c, d) in green, demonstrating the precision required to complete the tasks.

TABLE III
INFLUENTIAL FACTORS IN ROBOT COMPLETION TIME

Factor	Experimental (R)	Control (R)
Throughput	-0.82	-0.75
Completion Time	+0.70	+0.66
Path Efficiency	-0.61	-0.61
Delay	-0.16	-0.41
$\hat{\mathbf{W}}$	-0.28	–

in Table III. For both groups, the only significant correlation is with throughput ($p < 0.05$). Completion time and path efficiency at the end of session 2 are moderately correlated with final completion time ($p < 0.1$). In contrast, the delay between session 2 and session 3 is not correlated with the final completion time ($p = 0.71$), which suggests performance degradation is not a significant factor in completion times. The choice of $\hat{\mathbf{W}}$ is considered with respect to both activation component-wise and merged-component cosine similarity to the subject with significantly better control than any other subject. This produces only weak correlations ($p > 0.4$), indicating that the exact $\hat{\mathbf{W}}$ is not a significant influence. An analysis of electrode span and distance between activation components, expected to be the major factors influencing robustness, also gave weak correlations with performance ($p > 0.5$), likely due to variability across subjects. Future work will investigate these relationships more closely.

V. DISCUSSION

This work presents a novel motor learning-based control scheme capable of real-time simultaneous and proportional control of a large number of DoFs without requiring retraining between sessions. Using high density sEMG, a muscle synergy-inspired decomposition avoids the constraint on targeted electrode placement while maintaining the session-independent benefits shown in other motor learning-based control schemes. Despite not knowing either the input signals nor non-intuitive mapping, the subjects demonstrate learning trends consistent with typical motor skill learning. The same trends are seen with the control group using noiseless keyboard inputs, and are consistent with previous works which control fewer DoFs while targeting specific muscles [27]–[29], [31].

The proposed method is designed to be robust to small electrode displacements by approximating independent activation signals as mixtures of Gaussians, effectively introducing a spatial low-pass filter on noisy EMG while reducing them to more stable, low-dimensional inputs. Thus, robustness of the method is a function of electrode span within and distance between each activation signal. Future work will quantify robustness with respect to these two variables and measured electrode displacement, as in [51].

Two potential outliers were revealed in this study. One subject became confused in the middle of a session, and had to restart the exploration process to re-learn the controls. Another subject performed significantly better than the control group throughout the experiment. This subject finished session 3 in six minutes and reported the controls to be both intuitive and simple to use. As the only subject with significant video gaming experience, this supports findings that video gamers learn myoelectric control tasks faster [42], perhaps due to enhanced ability to explore the potential input space.

This study focuses on solely able-bodied subjects to validate the proposed method as a practical control scheme for general myoelectric interfaces. However, the concepts of motor learning and brain plasticity may also be applicable to amputees and subjects with muscular disorders [4]. These groups may not have voluntary control over muscles necessary for conventional intuitive controls with high functionality. By generating a $\hat{\mathbf{W}}$ which maximizes the number of independent activation signals, assistive myoelectric devices might offer a sense of intuitive control without requiring kinematic mappings [41], at the expense of the initial learning curve associated with motor learning. This will be explored in future work.

The performance of each subject with the VR interfaces correlates with the ability to control precision tasks with the robot, despite subjects feeling differences in the control scheme due to physical constraints. In particular, speed and accuracy, as measured by TP, significantly reflected their capabilities in the physical interface. This implies that VR may be used to implicitly train subjects to intuitively interact with a physical robot, as in [35]. It also suggests that VR can be used as an initial screening to determine the viability of a motor learning-based control scheme (i.e. overcoming the initial learning curve) for a potential user.

In summary, the proposed control scheme provides session-independent simultaneous and proportional control of myoelectric interfaces using muscle synergy-inspired inputs. These inputs reduce the constraint on exact electrode placement over muscles, and increase the potential functionality of motor learning-based control approaches. The controls can be enhanced simply by interacting with the interface, similarly to learning a new motor skill. The results confirm significant

learning trends correlating with a sense of more intuitive control, supporting this method as a viable technique for reliable long-term control of myoelectric interfaces.

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